Drug Misuse and Consequences

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ABSTRACT

Globally, drug misuse and its attendant consequences have become the bedrock of social ills, thus constituting a universal challenge plaguing many nations and causing concern to both Governments and the people.Despite this ugly scenario, the solution to drug misuse has remained enigmatic. This review aims to educate and provide assistance in alleviating and reducing the challenges of drug misuse and its consequences, particularly in Nigeria.A manual literature and internet (Medline, HINARI, PubMed and other databases) search were conducted. It was found that commonly misused drugshave a notorious property of addictive potentials and hence, the unique ability to perpetuate their use by compromising rationality and wreaking havoc on the faculty of self-preservation. This causes myriad of health challenges which inevitably leads to disorder, dysfunction or destruction. Addictive substances induce pleasant/pleasurable states (euphoria in the initial phase) or relieve distress. Continued use induces adaptive changes in the central nervous system (CNS) that lead to tolerance, dependence, sensitization (reverse tolerance), craving and relapse with attendant adverse physical, mental, social, psychological and economic consequences. Strategies to control drug misuse include reduction of harm, psychotherapy, drug treatment and prevention initiatives. Realistic media literacy for youth is imperative and religious grouping activities are protective. Remediations of misuse are directed at restoring the total wellbeing of the individual and the family together with improved political and socioeconomic conditions as well as effective prevention strategies.

Keywords: Drug, Misuse, Addiction.

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1. INTRODUCTION

The prevalence of some social ills like gangsterism, rape, prostitution, kidnapping, armed robbery, assault and assassinations in Nigeria in recent times has been attributed in most cases to the misuse of drugs. This situation is not peculiar to Nigeria because drug misuse is a universal challenge plaguing many nations of the world and causing concern to both Governments and the people. The therapeutic use of *drug* refers to a pharmacological preparation used in the prevention, diagnosis and treatment of an abnormal or pathological condition whereas the non-therapeutic use of *drug* commonly indicates the use of illegal or socially disapproved substances [1]. According to the World Health Organization (WHO) [2], a drug is "any substance or chemical that alters the structure or functioning of a living being". In the broadest sense, a drug is a chemical substance that has an effect on bodily systems, mood and behaviour. Also included here are a wide range of prescribed drugs and illegal as well as socially accepted substances.

Drug use refers to the inhalation, ingestion or injection etc, of a substance that is used for therapeutic purpose or as prescribed by healthcare professionals.*Drug misuse* is the use of a drug in a socially unacceptable way that is harmful or hazardous to the individual or others [3]. In short, it is the inappropriate use of a drug, usually the result of a psychoactive substance being consumed in a manner that is not recommended and hence making physical, social and psychological harm inevitable [4]. It may also represent the pattern of use: experimental, recreational and dependent.*Substance misuse or abuse* is often

used specifically to denote the misuse or abuse of alcohol and psychoactive drugs.

Drug abuse, often associated with addiction and dependence, is the popular term in the United States which refers to the use (usually by self-medication) of any drug in a way that deviates from the approved medical or social pattern, that is, the abnormal use of drugs for purposes other than their legitimate medical indications [1,4]. However, Government agencies regard the use of any illegal substance as drug abuse.

A manual literature and internet (Medline, HINARI, Pub Med and other databases) search showed that the phenomenal increase in the number of individuals with alcohol and other drug issues has attracted considerable concern as one of the most important public health challenges. This reviewwas undertakento educate and provide help in alleviating and reducing the challenges of drug misuse and its attendant consequences particularly in Nigeria.

2. Reasons for and commonly misused drugs

Some of the common reasons for drug misuse include peer pressure, the lure of popularity, mere curiosity, need for excitement and to feel good and happy, to be alert, reduce fatigue and to perform well in studies, to forget or overcome problems and stressful situations, to cope with low self-esteem or sadness, parental derivations and parental substance abuse, unemployment and environmental stressors, drug availability, high social class, poverty, poor mental health, perceived adult drug use, changing social values, cultism, violence, industrialization, urbanization, low religiosity, and demonic possession [4-7].

Nigerian and indeed African studies already cited, have indicated that the following drugs are commonly misused:

i. *Tobacco*is the most commonly reported substance use especially by students as a stress coping strategy. It is often used together with cannabis [5]

ii. *Alcohol (ethanol)* is a CNS depressant that when used causes hilarity, poor judgement, risk-taking behaviour and is associated with suicidal tendencies as well as severe withdrawal syndrome (delirium tremens, confusion, delusions, hallucinations, convulsions and cardiovascular collapse). It has taken its toll on individuals, families and society [6].

iii. *Cannabis* often smoked as a cigarette *("joint"),*is becoming increasingly popular because of the "high" (euphoria) reported by users. The main active component is tetrahydrocannabinol (THC). The effects of increased appetite, attenuation of nausea, decrease in intraocular pressure, bronchodilation and relief of chronic pain have led to the use of cannabinoids like dronabinol, nabilone, cannabidiol, in medical therapeutics. At present, there is no enough evidence to proof that cannabinoids can treat or lower the risk of cancer in people although different cannabinoids seem to have different salutary effects on various cancer types. However, the issue of cannabis use has continued to solicit controversy mainly because of the founded fear that cannabis may serve as a "gateway" to the consumption of "hard drugs" (just as tobacco and alcohol have done) and also for the alarm raised by the British Lung Foundation that the cancer risk of cannabis is underestimated [7-9].

iv. *Cocaine and crack* (cocaine base), when injected or inhaled, has immediate euphoric effects which include hyperstimulation, reduced fatigue and mental alertness. It is less misused because of cost and availability. Crack is smoked alone or on cannabis or tobacco. Cocaine is also used in combination with an opioid (heroin) – "speed balling" [10].

v. *Opioids(opiates, narcotics)* include opium, morphine, heroin, fentanyl, tramadol, and codeine usually present in cough syrups. After

an injection of morphine or heroin, users report feeling a surge of euphoria ('rush') and other effects. The danger of contracting human immunodeficiency virus (HIV) and hepatitis from injections is real [11].

vi. *Benzodiazepines*are CNS depressants that produce sedation and hypnosis, relieve anxiety and muscle spasm, and reduce seizures. Examples include *diazepam, bromazepam, chlordiazepoxide, lorazepam, rohypnol.*To get 'high', abusers use them with heroin or cocaine. Effects on the mind include amnesia, hostility, irritability and disturbing dreams. Rohypnol ("date rape" drug) is particularly misused to physically and psychologically incapacitate victims targeted for sexual assault by putting it in the beverage of unsuspecting victims. The drug leaves the victims unaware of what has happened to them [12].

vii. *Amphetamines (amphetamine and methamphetamine)* are stimulants used for treating narcolepsy and attention-deficit hyperactivity disorder (ADHD). Usually taken orally or injected, but the availability of crystallized methamphetamine ("ice") has promoted smoking as another route of administration. Chronic use produces paranoid psychosis that can result in homicide or suicide, hallucinations, violent and erratic behaviours [13].

viii. *Hallucinogens* include:

a. lysergic acid diethylamide (LSD) when used orally, "LSD trip" impairs one's ability to make sound judgements and see common dangers thus making the user susceptible to personal injury.

b.Ecstasy/MDMA (3,4-methylenedioxymethamphetamine) acts as both a stimulant and a psychedelic, a "party drug" commonly mixed with alcohol or cannabis and causes changes in perception including euphoria and increased sensitivity to touch, sensual and sexual arousal.

*c. Ketamine*is a dissociative anaesthetic that is abused for its ability to produce dissociative sensations, amnesia and hallucinations and hence has been used as a 'club drug' to facilitate sexual assault or ritual killings [12,14].

*ix. Inhalants*eg super glue, spray paints, petrol/gasoline, typewriter correction fluid, felt tip markers, air freshener, etc are invisible, volatile substances found in common household products that produce chemical vapours that are inhaled to induce psychoactive or mind-altering effects [15].

x. *Anabolic Steroids(testosterone, nandrolone, stanozol, methadienone, boldenone)* are misused in an attempt to promote muscle growth, enhance athletic or other physical performance, and improve physical performance when ingested, injected intramuscularly or applied to the skin usually at doses 10-100 times higher than the approved therapeutic doses. They can cause dramatic mood swings, increased feelings of hostility and aggression ("void rage") and impair judgement. Cessation may lead to severe depression and suicide, psychological dependence and addiction. Their adverse effects are many [12].

xi. S*ome drugs misused as abortifacients* (because of their pharmacological effects) in some parts of the world where provision of contraceptive services is poor or banned include:

- ***a.*** *Ethinyloestradiol /ethisterone combination (Menstrogen^{R})* can cause psychic depression leading to suicide.

- ***b.*** *Levonorgestrel (Postinor 2^{R})* which can cause heavy or prolonged haemorrhage.

- ***c.*** *Misoprostol*is misused because of its potential adverse effects of gynaecological disturbances (spotting, dysmenorrhoea, abortion etc).

xii. Nonsteroidal anti-inflammatory drugs (NSAIDs) are misused in Nigeria by unqualified prescribers who recommend 3 or more different NSAIDs together with a high dose of a corticosteroid such as prednisolone, to be taken as a single dose, to their unwary clients. This treatment option, which can be repeated several times in a day, is popularly known as "selection" and it is adjudged to be very efficacious in spite of its predisposition to dangerous adverse effects [16,17].

xiii. *Designer drugs (Legal "**highs")*** are drugs of abuse that are illicitly produced with the intent of developing substances that differ slightly from controlled substances in their chemical structure while retaining their pharmacological effects. They include herbal or synthetic substances eg.

a. *Mephedone*mimics the stimulant effect of cocaine or ecstasy (MDMA).
b. *Balt salts (synthetic cathinones)* is similar to cocaine, methamphetamines, ecstasy.
c. *K_2 and spice mimics* THC (cannabis)
d. *Synthetic opioids* mimic morphine, heroin, codeine, fentanyl. The sole purpose is to ensure these drugs do not fall under the Misuse of Drugs Act (UK) or the Controlled Substance Analogue Enforcement Act (USA).

xiv. *Drugs of concern* are those that despite their being uncontrolled at the moment, they pose great risks to individuals who misuse them eg;

a. *DXM (Dextromethorphan)* found in OTC cough mixtures as a suppressant is abused in high doses to experience euphoria and hallucinations.
b. *Kratom leaves* consumption produces stimulant effects (low doses), sedation (high doses), hallucinations, psychological and physiological dependence.
c. Salvia divinorum is a perennial herb abused for its hallucinogenic effects, causing psychic effects, anduncontrollable laughter [12].

3. Drug Misuse

Globally, drug challenges are as old as drugs. In recent years, the misuse of the non-therapeutic (licit/legal) drugs such as caffeine, tobacco/nicotine or alcohol as well as over-the-counter (OTC) drugs, prescribed medications and illicit/illegal psychoactive substances including cannabis, cocaine and crack, morphine, tramadol, codeine, heroin, benzodiazepines, etc, has increased dramatically, resulting in health and socioeconomic challenges, to which every healthcare professional must recognise and respond.

The notorious property of the above drugs is that they have addictive potentials and hence the unique ability to perpetuate their use by compromising rationality and wreaking havoc on the faculty of self-preservation. This causes a molecular brain disease which inevitably leads to disorder, dysfunction or perhaps destruction [18].

Drug addiction is a chronic relapsing disorder in which compulsive drug-seeking and drug-taking behaviour persists despite serious negative physical and psychosocial consequences. Addictive substances induce pleasant/pleasurable states (euphoria in the initial phase) or relieve distress. Continued use induces adaptive changes in the central nervous system (CNS) that lead to tolerance, dependence, sensitization (reverse tolerance), craving and relapse [18].

From earlier reports[19-21]as well as recent clinical and epidemiological surveys[5, 10-13, 22-25], it is evident that drug (substance) misuse is a growing major global public health issue. It was estimated that between 167 and 315 million people or 3.6 – 6.9% of the world adult population aged 15 – 64 years had used

an illicit substance in the preceding year. The global disease burden attributable to alcohol and illicit drugs is 5.4% while 3.7% is attributable to tobacco use alone. Of the misused drugs, the main ones used are opioids (narcotics, opiates) in European and Asian countries, cocaine in North America, cannabis in Africa and various combinations in the USA.

According to Emafo [21], the situation had become sufficiently disturbing such that the defunct Federal Military Government of Nigeria in a bid to remedy the situation enacted Decree 48 of 1989 which established the National Drug Law Enforcement Agency (NDLEA). The above reports have also demonstrated that drug misuse cuts across various age groups and social classes with the adolescents and young adults being in the majority. Global prevalence of drug use among College/University students is 12 – 85% whereas studies from Nigeria report 33 – 78% prevalence [4, 19, 20, 26-30].

*Risk factors for substance drug misuse*includeavailability of the drug; vulnerable personality type (one who is easily influenced and unable to face problems or challenges); poor educational system; poor student-teacher relationship; fanciful advertisements of alcohol and drug products in mass/social media; peer group pressure; urbanization; disadvantaged sections of society (low socio-economic status, high or low population density) and family factors (eg family history of drug use, dysfunctional backgrounds, weakening of family cohesion and of traditional values) [27,29].

*Possible signs of drug misuse include*frequent lying and evasive behaviour; irritability and quarrelsomeness with temper flare ups; keeping late nights; missing classes and poor academic performance; anduse of dark glasses usually to cover red eyes andreduce excessive light rays into over dilated pupils.Others are frequent demands for money (always "broke", always borrowing money); stealing and/or selling of personal items [31].

4. Consequences of Drug Misuse

Adverse consequences of drug misuse pose a threat to the Physical health, mental, social, psychological, and economic fabric of families, communities and nations [32,33].

i. Physical consequencesinclude liver diseases such as cirrhosis, fatty liver, liver cancer; infections including HIV/AIDS due to disinhibited risky behaviour while intoxicated; cardiovascular problems eg hypertension, heart failure; brain diseases eg Wernicke-Korsakoff syndrome, seizures, cerebral haemorrhage; malnutrition, weight loss and death from overdosage.

ii. Psychological consequencesinclude anxiety and restlessness; sleep disturbances; mood fluctuations; depression that may lead to suicide; apathy and lack of motivation (amotivational syndrome); delusions and hallucinations; low self-esteem and inferiority complex; psychosis, irrational talk that demonstrates that one is out of touch with reality.

iii. Mental consequencesinclude impaired cognitive functions like poor academic performance resulting in poor grades or outright withdrawal from school.

iv. Social consequencesinclude poor interpersonal relationships; insecurity in the community manifesting in armed robbery, violent crimes, rape, murders, kidnapping, hired assassinations etc.; prostitution, family breakdown causing stress for parents/siblings; traffic accidents and deaths due to driving under the influence of substances; arrests and imprisonments; and ultimately, failure to fulfil potentials and purpose of life.

v. Economic consequencesinclude loss of opportunities through dismissal or withdrawal from gainful employment; wastage of revenue that could have gone into savings and other productive projects; reduced productivity and loss of skilled manpower and personnel because sometimes, very talented and

otherwise highly intelligent and promising individuals are involved in drug misuse and addiction.

5. Strategies to Control Drug Misuse

Theseinclude reduction of harm, psychotherapy, drug treatment and prevention initiatives. However, prevention is the best strategy because when addiction sets in, recovery is a very challenging clinical goal [4]. Hence the urgent need to intensify the fight against the "gateway drugs" (tobacco and alcohol) cannot be over-emphasized. Realistic media literacy for youths is imperative. One who smokes tobacco or drinks alcohol is 65 times more likely to use cannabis than the one who abstains [25].

Practical preventive tipsinclude:

- Avoid looking for short cuts or 'escape routes' when faced with life challenges.
- Engage in problem-solving approach for overcoming challenges.
- Do not bow to peer pressure. 'Everyone is doing it,' is not an excuse.
- Religious groupings/activities are protective [32].
- Several people may not suffer any harm from it but it is not a risk worth experimenting with our own bodies/lives.

Indeed, in matters of drug misuse/addiction, prevention is always better and cheaper than cure!

6. Conclusion

Drug misuse is a major global public health issue. Complications of drug misuse can be managed by interventions directed at restoring physical, mental, psychological (emotional), social and spiritual health as well as family harmony together with improved political and socioeconomic conditions. Since prevention is the key to successful management, urgent attention should be paid to reduction of risk factors and strengthening protective factors against this menace by the international community, governments, mass media, institutions of learning, religious groups, NGOs, parents and teachers, and more importantly, healthcare professionals. Thus, by mass mobilization and by many means, we can effectively reduce and remedy the tragic effects of drug misuse and its consequences on individuals, families and the society.

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